RUBY CONF AFRICA

The Modern Rails Blueprint: A Practical Look at Hotwire and Solid Gems



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We are builders not bike-shedders!

Turbo Drive, Frames & Streams

Deep Dive into Turbo

Solid Gems

Service Objects, RubyLLm and Monads

Broadcasts

Stimulus JS

Building a chat application



Turbo Drive

This is responsible for handling requests. Turbo Drive intercepts all link clicks and replaces them with JS fetch requests. The same happens with form submissions or post requests.

The page body is then replaced with the response and the head objects from both pages merged. This is done to persist caching.

Can be disabled in the local scope by setting data-turbo attribute to false.

This can be seen on session and profile buttons

Turbo Frames

This is one of the most powerful tools in turbo that govern how reactive component are defined. This is how a page is decomposed into logical sections. Turbo frame tags allow you to gather and stitch together a new page based on components that are distributed in your views.

Turbo frames allow you to fetch only these logical blocks, defer loading, enable caching on components and get ready for mobile where pages are small and need to be fragmented as components.

Decomposed frames are also loaded parallel which makes pages load faster than creating series https bottlenecks.

Defined by turbo frame tags, loading and src attributes

Turbo Streams

Live actions require websockets(two-way) or server sent events(SSE - one way) which are both Transmission Control Protocols telling your device to communicate and stay in touch with a server.

Turbo streams builds on websockets to perform HTML Over the Wire (HotWire) communication by visiting the page where the turbo frame is located, rendering the frame, grabbing it's HTML content and broadcasting it over the websocket connection to our current page where we just display it!

This has three profound implications, no JSON walls, reuse of page components and existence of components.

Turbo stream tags include actions, targets and template formats.

Turbo Native is the last component for building native iOS and Android apps

Turbo Drive deep dive

01 We shall explore Turbo with confirm buttons, loaders and more. Turbo performs link preloads when hovered to make applications faster. This can cause side effects when hijacking events.

02 Turbo also fires events on form submission like turbo:submit-start, turbo:before-fetch-response etc that can be used to disable input forms, show loaders etc.

03 Besides the 7 main turbo actions we will cover, Rails 8 introduces turbo refreshes that are meant to work like broadcasts.

Turbo Frames Deep Dive

01 Turbo frames are denoted by their html tags that when rendered get html ids which must be unique. Rails provides a dom_id helper for this purpose

02 Turbo frames are meant to change how link and form navigation and caching/loading works. They are not connected to Turbo Streams and should not be used for that purpose which adds unnecessary complexity to your page

03 We shall explore eager loading, lazy loading and out of scope navigation with _top keywords

Coming to live with Turbo Streams

01 Turbo streams support 8 actions append, prepend, replace, update, remove, after, before, and refresh and a target attribute

02 A targets attribute can be used to act on multiple elements with targets='.class'

03 These events can be sent from the servers controller or via ActionCable layers as broadcasts. We shall explore both usages

Going Solid

01

Rails has had frameworks that worked to perform sidequest tasks that were either expensive in memory consumption or long running. These were often assigned to third party adapters like Redis/Memcache/Sidekiq and more.

02

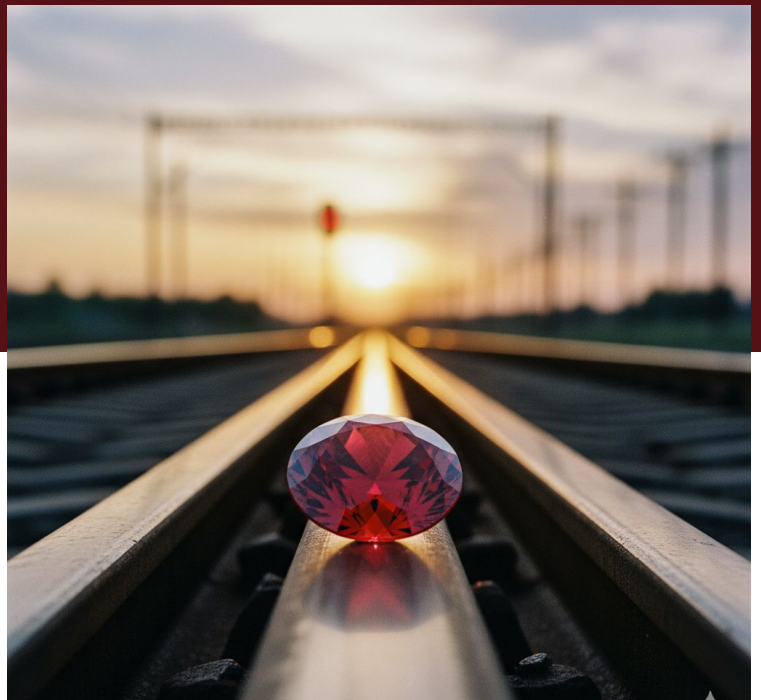
These adapters include ActiveJob which dispatched jobs to Sidekiq and Resque. This now works using Solid Queue in Rails 8+.

ActionCable dispatched messages and broadcasts to Redis and Memcache. This now runs on Solid Cable by default.

Finally, Solid Cache replaces Redis and other memory services to perform larger caching based on SSD/NVMe storage.

These services run their own parallel databases but you can choose to run them all on the default database

Stimulus



Stimulus is a component oriented JS library that takes the concept of decomposing your page features and running them on a separate JS controller class

Thank you!

Questions

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ARC meetups

